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From prompt engineering to modeling: secondary science teachers' use of generative AI to engineer sweetener molecules

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Scientific modeling is a core epistemic practice in chemistry, yet supporting learners in visualizing and making sense of abstract molecular structures and functions remains a challenge. While generative artificial intelligence (GenAI) offers new representational capabilities, it remains unclear how text-based prompt engineering intersects with established theories of scientific modeling. This study addresses this gap by reconceptualizing prompt engineering as a form of molecular modeling within AIMS, a chemistry-specific GenAI-supported molecular modeling environment. Through a qualitative case study, 11 secondary science teachers used GenAI embedded in AIMS to design novel sweetener molecules. Applying the CLEAR and REFINE analytic frameworks through the lens of Halloun's Modeling Theory, the study examined how participants translated disciplinary knowledge into iterative prompt-based molecular designs. The analysis concludes with three key findings. Participants' prompts functioned as externalized models that embedded assumptions about molecular structure–function relationships and design constraints within explicit chemical and theoretical requirements. As iterations progressed through model testing, reasoning shifted from broad appeals for scientific accuracy to precise, mechanistic constraints, including spatial arrangements and molecular properties. Revisions operated as controlled experiments where participants tightened or relaxed constraints to test the plausibility of GenAI outputs. These findings advance chemistry education by demonstrating that a chemistry-specific GenAI-supported modeling tool can function as a modeling assistant rather than an answer engine, scaffolding the core epistemic practices of construction, evaluation, and refinement in chemistry education. This reconceptualization positions prompt engineering as a source of written evidence for how learners articulated, tested, and revised chemical constraints, offering educators new ways to support and assess student modeling practices.

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Introduction

Scientific modeling is a core epistemic practice in science education that emphasizes the construction, evaluation, and refinement of representations that explain and predict natural

phenomena (Gilbert, 2004; Hestenes, 2010; NGSS Lead States, 2013). In this study, we follow Halloun (2006) and operationalize modeling as an iterative process in which learners generate representational artifacts (*e.g.*, verbal descriptions, structured prompts, or molecular sketches), test them against evidence, and revise them toward greater explanatory and predictive coherence. Prior work shows that engaging learners in such practice supports conceptual understanding and scientific reasoning across grade levels (Bamberger and Davis, 2013; Forbes *et al.*, 2015; Roth *et al.*, 2020; Malone and Schuchardt, 2023).

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Modeling is central to chemistry education, yet it remains challenging. To reason chemically, learners must move across multiple levels of representation while coordinating spatial relationships, functional roles, and disciplinary norms of plausibility (Gkitzia *et al.*, 2020; Talanquer, 2022). These demands are particularly pronounced in molecular design tasks, where learners must consider how specific structural features and spatial configurations give rise to target properties. The present study examines this challenge in the disciplinary context of designing low- or noncaloric sweetener molecules. This context is instructionally productive because it requires learners to connect molecular structure and function with a theoretical account of sweetness, such as the AH-B-X theory, which explains sweet taste in terms of a hydrogen-bond donor site (AH), a hydrogen-bond acceptor or basic site (B), and a hydrophobic site (X) arranged in a complementary spatial relationship to support interaction with the sweet taste receptor protein (Zheng *et al.*, 2024).

Because chemistry modeling depends heavily on making ideas visible, manipulable, and testable, prior studies have explored representational supports that help learners externalize and refine their thinking. One line of work explores student-generated animations. Berg *et al.* (2019) found that storyboarding, physical modeling, and stop-motion animation created structured opportunities for learners to articulate decisions about scale, particle interactions, and representational choices, which in turn supported more coherent and scientifically grounded explanations. Other studies have focused on the use of physical models as feedback. Padalkar and Hegarty (2015) demonstrated that aligning students' drawings with concrete molecular models reduced spatial errors and improved diagram-translation accuracy more effectively than verbal feedback alone. Similar benefits have been observed in model-based workshop settings. Stieff *et al.* (2016) found that students who participated in constructing and translating molecular representations demonstrated stronger representational competence and spatial reasoning. A unifying thread across these studies is the structured opportunity they provide for learners to externalize and iteratively refine their thinking through interaction with representational tools.

Recent advances in generative artificial intelligence (GenAI) introduce new possibilities for supporting these processes within AI-supported learning contexts where learners engage through prompts as multimodal inputs (Tassoti, 2024; Akaygun and Kilic, 2025). GenAI can serve as an interactive space where learners articulate their understanding through prompts, receiving feedback as a structured response. Emerging research on prompt design suggests that these chat-based comments can function as epistemic tools by specifying key components

such as persona, aims, recipients, themes, and output formats (Lo, 2023; Park and Choo, 2025). Through these elements, prompts encode users' intentions, establish constraints, and externalize reasoning, creating traceable records of how ideas evolve across prompt histories and outputs. However, little empirical work has examined how these textual interactions correspond with established theories of scientific modeling or how they might be harnessed to support disciplinary learning. In a learning context, educational value emerges from both automated generation and how learners formulate, test, and refine their ideas through iterative prompt engineering, mirroring the construction-evaluation-revision cycles that characterize effective modeling-based instruction.

To study this process in the context of molecular sweetener design, we used AIMS (Artificial Intelligence for Molecular Sciences), an open-source, browser-based molecular modeling platform designed to foster molecular reasoning and computational thinking (Institute for Future Intelligence, 2025). AIMS is chemistry-specific and GenAI-supported, allowing users to generate novel molecules from text-based prompts, inspect and compare generated molecules, analyze their chemical and physical properties through built-in visual analytics and molecular dynamics simulation (Fig. 1). Because the platform also archives prompts and generated molecules within a project, it provides a traceable record of how users iteratively generate, evaluate, and revise molecular designs. These features position AIMS as a human-AI molecular modeling environment in which prompt-based interaction can externalize, test, and refine learners' conceptual understanding through iterative design in ways that parallel the process of inquiry-driven scientific modeling.

Although prior work has documented a range of representational interventions (Padalkar and Hegarty, 2015; Stieff *et al.*, 2016; Berg *et al.*, 2019) and, separately, the affordances of prompt design (Lo, 2023; Park and Choo, 2025), their intersection remains largely unexplored, particularly in teacher learning contexts. Teachers occupy a critical position in this emerging landscape as professionals who must interpret, adapt, and translate new technologies into meaningful instructional practice (Bereczki and Kárpáti, 2021; Frawley and Campbell, 2025). Understanding how teachers externalize disciplinary knowledge, evaluate AI-generated outputs, and iteratively refine their reasoning is therefore essential for designing effective professional learning experiences. Advancing this understanding requires studies that examine these processes with analytic rigor, ground them in established theoretical frameworks, and identify strategies to make modeling more accessible and teachable in GenAI-supported settings.

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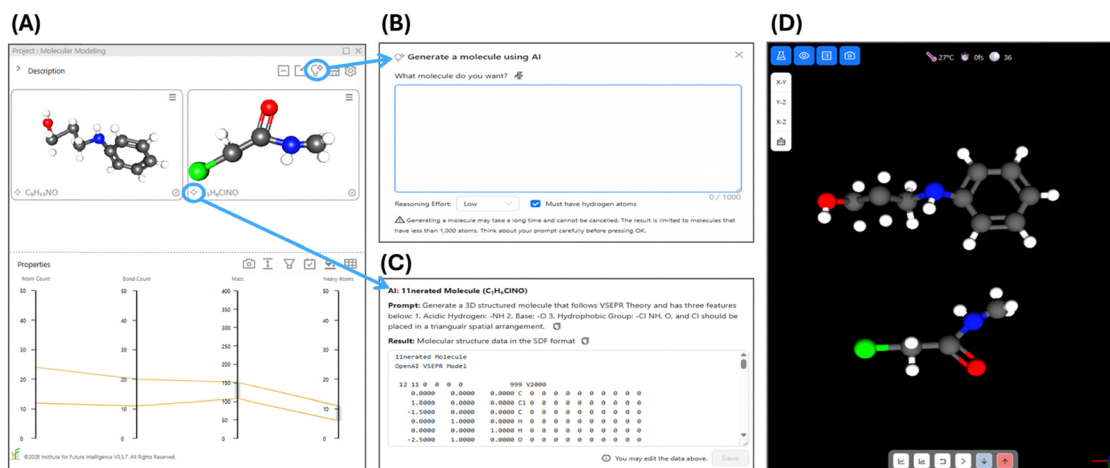


Fig. 1 GenAI-supported molecular modeling environment in AIMS. Note. (A) Molecule gallery for storing, comparing, and inspecting generated molecules and their basic chemical and physical properties. (B) Prompt interface where users specify text-based requests for new molecules. (C) Copyable text area that archives each prompt and the corresponding result from GenAI. (D) Molecular dynamics simulation window for visualizing, comparing, and animating 3D molecular structures.

These strands of work point to the need for empirical studies that examine how GenAI-supported modeling unfolds in concrete disciplinary contexts and how teachers, in particular, engage with these tools as part of their professional learning. This study contributes to chemistry education research by conceptualizing prompt engineering as a form of molecular modeling, empirically documenting how secondary teachers translate design goals into molecular constraints such as functional groups and spatial relationships, and providing analytic tools for examining prompt histories as traces of chemical reasoning. To examine these processes and guide our analysis, we draw on Halloun's (2006) Modeling Theory and the Modeling Learning Cycle (MLC), which together frame prompt construction and revision as forms of scientific modeling that can be systematically examined in AIMS.

Halloun's modeling theory

This study is grounded in Halloun's (2006) Modeling Theory and its corresponding MLC, which together provide a lens for understanding science learning as a process of model construction, evaluation, and refinement. Halloun conceptualizes science learning as a dynamic interaction between the physical world, comprising observable phenomena, and the mental world, where learners construct and organize representations of those phenomena. Within the mental world, individuals develop conceptions of how the world works. Through instructional and experiential interactions, learners reorganize their conceptions into coherent and scientifically viable models, which are structured conceptual systems that represent aspects of physical reality. In this view, models serve as both the means and products of conceptual understanding, illustrating the degree to which a learner's ideas accurately represent physical phenomena.

Halloun's MLC operationalizes this epistemological perspective by describing how learners develop and refine models through iterative reasoning. Although the MLC comprises five phases, this study focuses primarily on the two central phases, *Model adduction* and *model formulation*. These phases most

directly align with the study's focus on how participants articulated molecular ideas in prompts and then revised those prompts in response to generated outputs during human-AI interaction.

In Model Adduction, learners propose one plausible model that appears to best account for a newly encountered pattern or phenomenon. This model is formulated based on learners' existing conceptions, theoretical understanding, and available evidence. Model adduction therefore represents the stage in which learners shift from multiple tentative ideas toward a single, coherent representation.

Model Formulation follows, in which learners refine and elaborate their tentative model through systematic investigation and analysis. Empirical evidence or logical reasoning is used to assess the adequacy of the model, leading to successive revisions that increase its explanatory or predictive power. Learners evaluate relationships among model components, identify missing components, and incorporate necessary concepts or principles to strengthen its structure. Through this process, the model evolves toward a more scientifically viable form.

In the context of this study, MLC provides a foundation for interpreting how learners engaged in modeling practices while using GenAI to design molecular sweeteners. Here, a prompt is treated as a written model that externalizes part of a learner's conceptual understanding in a natural language form. Each prompt encodes the participant's conceptions about molecular properties, spatial arrangements, and theoretical alignments, such as those described by the AH-B-X theory of sweetness (Zheng *et al.*, 2024). Through cycles of generation and refinement, prompts become manifestations of model construction and revision, revealing how conceptual understanding is articulated, tested, and refined in a generative environment. Within this framing, Halloun's Modeling Theory explains why prompt engineering can be understood as a modeling practice and prompts as written models, whereas the CLEAR and REFINE frameworks provide the analytic lens for examining the quality of those prompts and their revisions.

Prompt characteristics: CLEAR and REFINE

To examine participants' prompt-based modeling systematically, two complementary analytic frameworks were employed: CLEAR and REFINE. These frameworks, adapted from Lo (2023) and Park and Choo (2025), provide a structured approach to interpreting participants' modeling behaviors as expressed through their written prompts and subsequent revisions. CLEAR characterizes the structure of prompts, while REFINE captures the process of prompt revision. These frameworks offer a coherent lens for analyzing how participants constructed and refined prompts as forms of scientific modeling.

CLEAR includes five dimensions of effective prompt design: Concise, Logical, Explicit, Adaptive, and Restrictive. *Logical* represents the sequential organization of information that enables a stepwise reasoning process. *Explicit* involves the use of precise and specific language that minimizes ambiguity through the inclusion of detailed parameters or examples. *Adaptive* reflects the contextual tailoring of prompts to align with specific goals, audiences, or disciplinary contexts. Finally, *Restrictive* defines boundary conditions such as scope, length, or representational format. The Concise dimension was excluded due to its subjective interpretation.

While CLEAR focuses on the structure, REFINE describes the strategies learners use to improve prompts through iterative reasoning. *Rephrase* refers to modifying word choice or phrasing. *Experiment* involves altering contextual details or adding examples to refine the focus of the prompt. *Feedback* captures recursive interactions that build upon prior exchanges to sustain continuity and contextual relevance. *Inquiry* represents the use of targeted questions to verify, probe, or challenge responses. *Navigate* reflects the use of trial-and-error cycles through successive prompt-output iterations. Finally, *Evaluate* refers to critically assessing generated outputs for accuracy, feasibility, and relevance. In this study, the GenAI system's lack of responsive feedback limited the presence of Feedback and Inquiry dimensions; therefore, the analysis primarily focused on Rephrase, Experiment, Navigate, and Evaluate as indicators of revision strategies.

Together, modeling theory, CLEAR, and REFINE establish the foundation for examining how teachers engage in modeling with GenAI. Modeling theory provides a lens for understanding modeling as an iterative process of constructing, evaluating, and refining conceptual representations (*i.e.*, models), while CLEAR and REFINE operationalize how these processes become visible in participants' written interactions.

The present study

Although prior research has examined modeling in science education and prompt design in AI-supported learning environments, there is a growing need to understand how teachers engage with modeling practices, especially in relation to three-dimensional (3D) molecular design. This study conceptualizes prompt engineering as a form of molecular modeling and empirically examines how secondary science teachers translate design goals into prompt structures, evaluation criteria, and revisions when engineering sweetener molecules. Doing so

illuminates prompt-based interaction as collaborative modeling practice in chemistry education. The study is guided by two research questions:

- RQ1. How do participants engage GenAI through design goals, prompt construction, evaluation, and iterative revision to model candidate sweetener molecules?
- RQ2. How do participants justify the selection of their final sweetener molecule?

Methods

Study design and participants

This study employed a qualitative case study design to examine secondary science teachers' engagement in a GenAI-supported molecular modeling task (Clark and Creswell, 2015). The case was bounded by time, setting, activity, and participants: the final week of a four-week professional learning program, the AIMS-supported GenAI sweetener design activity, and program participants. Although the broader professional learning program provided the instructional context, the primary analytic focus was the final-week GenAI sweetener design task, which generated the prompt logs, evaluation notes, revision records, and final justifications analyzed in this study. The case was selected because it represented a novel and contextually rich instance of teacher engagement in GenAI-supported modeling, which is an emerging area in science education where little empirical research exists.

A purposeful sampling strategy (Palinkas *et al.*, 2015) was used to recruit participants who met the case criteria. Eleven teachers from four U.S. states were invited to participate through a written and oral description. All eleven (100%) consented, provided informed consent, and became research participants. Research participation was voluntary, but completion of the professional development program was compensated with a stipend. All research participants (henceforth, participants) were certified teachers with established disciplinary expertise. Five reported more than 16 years of teaching experience, four reported 11–15 years, and two reported 0–10 years. The sample was predominantly female (71% female, 29% male), and the study was approved by an Institutional Review Board review (IRB # 202400747).

The first three weeks of the program were not the primary site of analysis; rather, they provided shared conceptual preparation for the focal GenAI modeling task. During these weeks, participants completed a sequence of modeling-based lessons using the AIMS platform that progressively introduced foundational concepts: cheminformatics basics (week 1), molecular structure and function (week 2), and the AH-B-X theory of sweetness (week 3). In week 3 participants completed a modeling activity that explored the molecular basis of sweetness without GenAI support. They began by analyzing commercial product labels to classify six common sweeteners as caloric or non-caloric, then used AIMS to examine their molecular structures, identifying elemental composition, functional groups, and structural features. From this analysis, participants constructed initial

explanatory models (*i.e.*, diagrams and explanations) of how molecular features relate to sweetness and caloric content. Using the AH-B-X theory of sweetness, they refined these models and applied them to a baking scenario in which they evaluated new sweeteners and justified their recommendations. This preparatory sequence established a shared conceptual foundation in structure–function relationships, sweetness mechanisms, and caloric content.

In the final week, participants completed the GenAI artificial sweetener design task, which served as the core of this case. Their engagement in this task was shaped not only by the GenAI environment, but also by the preparatory activity introduced in the earlier lessons. Within AIMS, they were instructed to design a sweetener molecule using the embedded GenAI feature by writing and revising prompts in response to the system's output. Output from GenAI was in the Structure-Data File (SDF) format, which was interpreted and presented to the user as an interactive 3D molecule in AIMS, with the predicted chemical and physical properties displayed in a parallel coordinate plot used to visualize high-dimensional data. The worksheet directed them to record each prompt, the corresponding output, and their evaluation and revision strategy, guiding them through a structured modeling process aligned with MLC.

Activity description

The activity unfolded in three phases (Table 1). First, participants indicated their design goals and created an initial GenAI prompt to generate a candidate molecule (Fig. 2). This stage corresponded to Model Adduction, where participants proposed plausible models based on their understanding of sweetener properties and structure–function relationships. Once an initial molecule was generated, they used AIMS analytical tools (*e.g.*, data visualization and molecular dynamics simulations) to evaluate the output and identify refinements needed. Participants then revised their prompts and generated new candidate molecules, repeating this process multiple times until satisfied with the molecule generated. Participants recorded each prompt, its evaluation, and their prompt strategy in a chronological prompt log, creating a traceable record of how their thinking and strategies evolved. These iterative refinements represented Model Formulation, as participants empirically tested their models and applied disciplinary reasoning to progressively improve prompt specifications. Finally, once satisfied with a generated molecule, participants

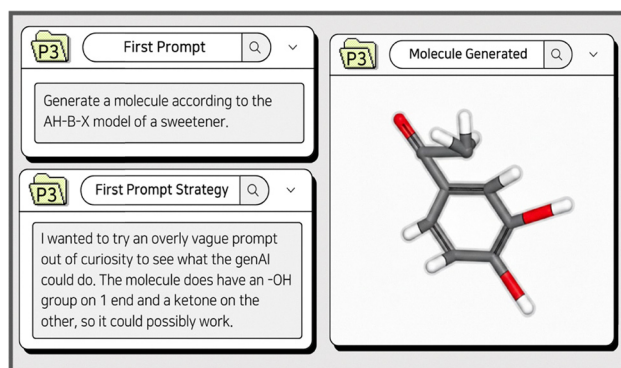


Fig. 2 Example of an initial GenAI prompt, strategy, and output. Note. The label “P3” refers to Participant 3. The top-left panel shows the participant’s first prompt to GenAI, while the panel directly below it contains the participant’s explanation of their prompt strategy. The right panel displays the molecule that GenAI generated in response.

finalized their prompt and composed a written rationale explaining their selection (Fig. 3). This justification phase required participants to articulate how their final molecule aligned with both molecular evidence and their design goals.

Measures and data collection

Multiple forms of qualitative data were collected. Participants’ worksheets served as the primary data source, containing design goals, prompt and strategy logs, evaluation notes, and justifications for final molecules (Table 2). These documents provided evidence of participants’ reasoning and modeling behaviors.

Data analysis

All data were analyzed through qualitative thematic analysis (Braun and Clarke, 2006) using MaxQDA 24.2 (VERBI Software, 2024). The analysis involved both descriptive and interpretive phases to identify how participants constructed, evaluated, revised, and justified GenAI-generated molecules. To address different aspects of this process, we used two units of analysis: the participant and the prompt. This dual-level approach was necessary because the number of iterative cycles varied substantially across participants, ranging from zero to ten subsequent prompts. As a result, relying only on prompt-level analysis would have disproportionately weighted participants who revised more frequently and could have obscured patterns present across the cohort as a whole.

Table 1 Alignment between MLC and the GenAI molecule design activity

MLC phase	Definition (Halloun, 2006)	Operationalization in the activity
Model Adduction	Learners propose one plausible model that appears to be reliably mapped onto the new pattern being investigated in the cycle	Participants identified design goals for their sweeteners, specifying desired molecular properties and structural patterns. They then constructed and submitted an initial prompt to generate a candidate molecule, externalizing their conceptions in a language model (<i>i.e.</i> , a prompt)
Model Formulation	Learners gradually formulate a comprehensive and scientifically sound model based on empirical testing and rational deduction	Participants evaluated generated molecules using AIMS analytical tools, assessing chemical plausibility and alignment with the sweetness theory. They iteratively revised their prompts to refine structural and functional specifications until satisfied, then justified their final prompt and selected molecule as the end result

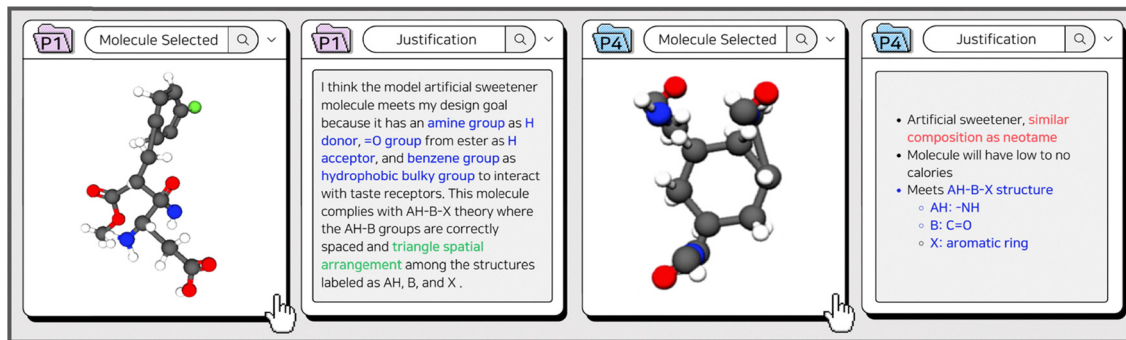


Fig. 3 Examples of participants' final molecules and justifications. Note. The labels "P1" and "P4" refer to Participant 1 and Participant 4. Color-coded text in the justification panels distinguishes different types of reasoning: blue identifies functional-group-related evidence, green references to spatial arrangement or positioning, and red highlights specific examples participants used to substantiate their reasoning.

Table 2 Research questions, data types, and analysis methods

RQs	Focus of analysis	Data type	Analysis method
RQ1.	Design goals and features of initial prompts	Design goals First prompts	Inductive thematic analysis Deductive coding using CLEAR and inductive thematic analysis
	Reasoning used to assess GenAI-generated molecular outputs Prompt revision strategies and features of revised prompts	Evaluation notes Prompt and strategy logs	Inductive thematic analysis Deductive coding using CLEAR/REFINE and inductive thematic analysis
RQ2.	Justification for final molecule selection	Justifications and images of molecule selected	Inductive thematic analysis

The participant level was therefore used for aspects of the task that reflected participants' overall reasoning across the activity. Specifically, design goals, evaluation strategies, revision strategies using REFINE, and final rationales were coded at the participant level and reported the proportion of participants ($n = 11$). This allowed us to show how widely particular forms of reasoning were represented across the sample while avoiding overrepresentation of participants who produced many more prompt iterations than others.

The prompt level was used to examine how the prompts themselves changed across iterative cycles. Each prompt ($n = 11$) and each subsequent prompt ($n = 30$) was coded using CLEAR and inductive subthemes for Explicit features. This prompt-level analysis allowed us to trace changes in the structure, specificity, and complexity of prompts over time.

To ensure analytic rigor, two researchers first reviewed a subset of responses to familiarize themselves with the data and refine a shared codebook. The codebook combined the *a priori* CLEAR and REFINE dimensions used to characterize prompts and revisions with inductively developed codes for characteristics of design goals, evaluation, and rationales. They then independently coded 20% of the dataset, consistent with recommended practice for reliability checks (O'Connor and Joffe, 2020). Intercoder reliability indicated substantial agreement between coders (Cohen's $\kappa = 0.67$) (McHugh, 2012). Coding disagreements were discussed and resolved through consensus, after which the first author coded the remaining dataset. Operational definitions of the CLEAR and REFINE dimensions used in this study are summarized in Table 3.

Findings

This section presents findings aligned with two research questions examining how participants engaged GenAI as part of a molecular modeling process and how they justified their final candidate sweetener. Findings are organized around participants' design goals, initial prompts, evaluation practices, and iterative revisions (RQ1), followed by their justification and selection of a final molecule (RQ2). The findings characterize GenAI use as embedded within modeling practice not as an extrinsic intervention.

Design goals

Participants' design goals established the initial boundaries for their GenAI-supported molecular design work (Table 4). The most common goals focused on *chemical and structural specifications*. Chemical feature constraints ($n = 5$, 45%) most often included functional groups and, in some cases, details about aromaticity, molecular size, or carbon count. Theoretical anchoring in the AH-B-X theory was also common ($n = 5$, 45%), indicating that many participants began from an established theory, not just relying only on general chemical knowledge. Scientific accuracy and validity ($n = 2$, 18%) included aspects such as plausibility, caloric or non-caloric designation, and 3D representation. Reference to a known sweetener, aspartame, appeared less frequently ($n = 1$, 9%) and was used to support the plausibility of their designs.

Beyond chemistry, a smaller number of participants described how they planned to use GenAI. Iterative prompting ($n = 2$, 18%)

Table 3 Operationalization of CLEAR and REFINE

Framework	Dimension	Operationalization in this study
CLEAR	Logical	Prompt organizes multiple requirements into a sequenced multi-clause request
	Explicit	Prompt specifies chemical, structural, or theoretical requirements for the molecule
	Adaptive	Prompt situates the request within the artificial sweetener design task
REFINE	Restrictive	Prompt sets firm boundaries that limit the solution space for GenAI
	Rephrase	Revises a prompt by clarifying or specifying language while maintaining the same underlying chemical and structural constraints
	Experiment	Treats revisions as controlled trials by systematically manipulating context or constraints to observe how outputs change
	Navigate	Uses iterative generate-evaluate-revise cycles
	Evaluate	Evaluate each output using chemical properties and theory

Note: the concise dimension was excluded from analysis due to its subjective interpretation, and the feedback and inquiry dimensions were excluded due to the system's lack of responsive feedback.

involved refining prompts across multiple cycles until the generated molecule met their expectations. Knowledge calibration ($n = 2$, 18%) reflected participants' intentions to probe or guide the system's chemical understanding, such as testing whether it recognized terms like "acidic hydrogen" or adding constraints to shape its responses. The themes suggest that design goals functioned as both the molecular target participants pursued and the strategy they expected to use with GenAI.

Initial prompts

When participants translated their design goals into initial prompts, these statements exhibited distinctive patterns across the CLEAR dimensions, with *Explicit* being the most dominant. Explicit features appeared in all initial prompts ($n = 11$, 100%), with participants articulating detailed chemical, structural, or theoretical requirements for the target molecule, often naming specific functional groups, AH-B-X alignment, or anchoring the design to a known sweetener. *Adaptive* features were also common ($n = 9$, 82%), typically situating the request explicitly within an artificial sweetener design context and signaling properties such as noncaloric status or intended use.

Restrictive features were present in seven initial prompts ($n = 7$, 64%) and set firm generative boundaries, particularly related to three-dimensionality and scientific accuracy. *Logical* structuring appeared less frequently ($n = 4$, 36%), where participants

organized multiple constraints into ordered, multi-clause instructions rather than a single sentence. These patterns indicate that participants' initial prompts reflected modeling intent by specifying structural constraints, functional relationships, and evaluative criteria.

Because the Explicit dimension encompassed multiple types of requirements, inductive thematic coding was conducted to unpack it into chemically and theoretically meaningful categories. This analysis identified five themes with associated subthemes (Fig. 4). Across the 11 initial prompts, *Theory*, *Feasibility*, and *Validity Anchors* emerged as a central theme. Just over a quarter of participants explicitly referenced the AH-B-X sweetness framework ($n = 3$, 27%), and more than half required the molecule to be scientifically accurate or feasible ($n = 6$, 55%). These anchors show that participants coupled generative exploration with explicit expectations for theoretical alignment and chemical plausibility.

Functional Groups and Interaction Roles provided a second major emphasis. Functional group targeting appeared in six prompts ($n = 6$, 55%), with additional attention to hydrogen-bonding interactions ($n = 1$, 9%) and acid-base features ($n = 1$, 9%). In these cases, participants specified particular functional roles they expected the molecule to play (for example, groups that could donate or accept hydrogen bonds), indicating that initial prompts often attempted to constrain GenAI by

Table 4 Themes of participants' design goals and prompt engineering strategies

Theme	Excerpt	<i>n</i> (%)
(1) Chemical and structural specifications		
(1-1) Chemical feature constraints	"I would like to set a certain number of carbon atoms , following the functional groups according to the AH-B-X theory" (P5)	5 (45)
(1-2) Theoretical anchoring	"I will use the GenAI feature to create a novel non-caloric sugar molecule following the AH-B-X sugar structure " (P8)	5 (45)
(1-3) Scientific accuracy and validity	"I intend to create a 3-D model of a novel artificial, noncaloric sweetener that is scientifically accurate and contains functional groups and aromatics" (P6)	2 (18)
(1-4) Reference to known sweeteners	"I intend to use GenAI tools to design a new artificial sweetener that has structural similarity to aspartame , follows the AH-B-X theory..." (P1)	1 (9)
(2) Prompt engineering and AI interaction strategies		
(2-1) Iterative prompting strategy	"I'll likely prompt it to recall many of the previous homework's non-caloric sweeteners as a base and then tweak it with successive prompts until I am satisfied " (P11)	2 (18)
(2-2) Knowledge calibration	"I will start with the broad model. Depending on how that works , I will provide more specificity... I'm curious to see if the GenAI knows what an "acidic hydrogen" means (for example) based on what it has already been taught" (P3)	2 (18)

Note: percentages indicate the proportion of participants ($n = 11$) who used each rationale at least once.

Explicit Dimension Subthemes (Initial Prompts)

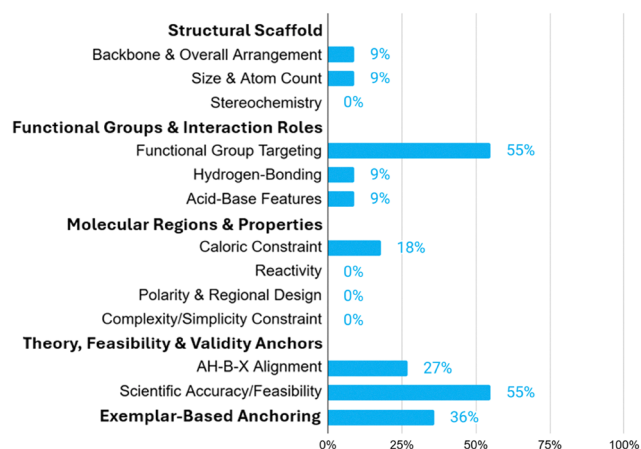


Fig. 4 Characteristics of initial prompts across explicit subthemes.

foregrounding interaction mechanisms relevant to sweetness potential.

Other themes appeared less frequently but further illustrate the range of explicit strategies. *Structural Scaffold* features, such as specifying backbone arrangement or atom-count constraints, each appeared in one prompt ($n = 1$, 9%). *Molecular Regions and Properties* were reflected in two prompts ($n = 2$, 18%), primarily through explicit caloric constraints on the designed molecule. *Exemplar-Based Anchoring* appeared in four prompts ($n = 4$, 36%), where participants referenced known sweeteners, such as aspartame and sucralose, to guide or benchmark the design. These themes show that even initial prompts encoded layered chemical, structural, and theoretical considerations to shape GenAI's first-round molecular outputs. Illustrative excerpts for each subtheme are included as SI.

Evaluation of generated molecules as model testing

After generating molecules, participants evaluated outputs using both chemical and theoretical criteria (Table 5). The most common evaluation strategies involved testing chemical

properties ($n = 4$, 36%) and theory-based reasoning ($n = 4$, 36%). Participants frequently identified unrealistic or unusual molecular configurations, such as distorted or implausible structures (this phenomenon, pertinent to structure generation using LLMs, is known as spatial hallucination), missing functional groups, excessive electronegativity, or limited rotatable bonds. Note that this article only reports the results of our study with a mainstream GenAI model in 2025 and these problems may be resolved over time when GenAI models become more powerful in chemical reasoning.

They also used the AH-B-X theory to assess whether generated molecules included plausible hydrogen donors, acceptors, and hydrophobic groups related to the sweetness potential. Scientific accuracy was also a focus for several participants ($n = 3$, 27%), who questioned whether the generated molecules were realistically represented in 3-D or chemically feasible. In contrast, comparisons to existing sweeteners were rare ($n = 1$, 9%), with only one participant referencing neotame as a plausibility check.

These findings show that participants engaged in systematic and critical evaluation of GenAI outputs. Evaluation operated as a form of model testing, where AI-generated molecules were judged against participants' conceptual understanding. Instead of accepting outputs as complete or authoritative, participants treated them as artifacts to be interrogated, applying their disciplinary knowledge to judge plausibility, coherence, and alignment with established chemical principles. Evaluation therefore served as the basis for subsequent prompt revision.

Iterative revision as model refinement

Across successive prompt-output cycles, each participant generated between zero and ten subsequent prompts, with 30 subsequent prompts produced in total. One participant achieved their design goal using only the initial prompt. We analyzed these revisions using REFINE to characterize the revision characteristics and through CLEAR to describe how the prompt structure evolved over time. The findings reflected shifts in modeling assumptions and constraints, suggesting

Table 5 Distribution of evaluation strategies and representative examples from participants

Theme	Example	<i>n</i> (%)
(1) Chemical properties		
(1-1) Structure-based assessment	"I got a branched hydrocarbon with 4 hydroxyl groups , but the 3-D structure looked weird (straight, not bent)" (P7, 2nd evaluation)	4 (36)
(1-2) Functional group related assessment	"This prompt did not generate a benzene ring although stated in the prompt" (P8, 2nd evaluation)	2 (18)
(1-3) Electronegativity	"The model spit out something with Fluorines which I think are too electronegative " (P10, 1st evaluation)	1 (9)
(1-4) Distance between certain atoms	"It also included chlorines again too close together " (P11, 3rd evaluation)	1 (9)
(1-5) Lack of rotatable bonds	"It also looked like it lacked many rotatable bonds " (P11, 3rd evaluation)	1 (9)
(2) Theory reference	"The model artificial sweetener has an amine group as H donor , =O group from ester as H acceptor , and benzene group as hydrophobic bulky group to interact with taste receptors" (P1, 2nd evaluation)	4 (36)
(3) Scientific accuracy	"The previously generated molecule appeared in 2D, which does not reflect how molecules exist in real life " (P4, 1st evaluation)	3 (27)
	"I'm not sure that it is scientifically accurate " (P9, 1st evaluation)	
(4) Reference to existing sweeteners	"This gave me a similar compound to neotame " (P4, 5th evaluation)	1 (9)

Note: percentages indicate the proportion of participants ($n = 11$) who used each evaluation strategy at least once.

that participants' revisions reflected an evolving understanding of both their molecular design criteria and the capabilities of the GenAI-supported AIMS environment.

Revision strategies (REFINE). Within REFINE, revision strategies were summarized at the participant level to indicate how many participants used each type of strategy at least once, whereas changes in prompt structure across iterative cycles were examined separately at the prompt level through CLEAR. Within REFINE, the most widely observed approaches across participants were *Navigate by Iterations* ($n = 10$, 91%) and *Evaluate Outputs* ($n = 10$, 91%; Table 6). In practice, this meant that participants repeatedly generated a molecule, inspected the GenAI output for chemical and theoretical adequacy, and then adjusted their next prompt in light of that testing. *Experiment with Context, Examples, Constraints* was also common ($n = 9$, 82%), with participants treating revisions as controlled trials in which specific conditions were added, relaxed, or reconfigured to observe how the output changed as a result.

Within the *Experiment with Context, Examples, Constraints* dimension, three related constraint-based strategies emerged. *Constraint tightening* ($n = 6$, 55%) involved adding or sharpening requirements to increase chemical precision or strengthen alignment with the AH-B-X framework. For instance, after noticing that a generated molecule lacked an electronegative group that could function as a base, P5 explicitly required such a group in the next prompt. Both *tightening and relieving* constraints ($n = 2$, 18%) reflected altering between more and less restrictive prompts across successive iterations. For example, P7 iteratively modified an erythritol-based structure by first requiring both a benzene ring and a hydrogen bond acceptor site, then temporarily dropping the acceptor requirement, and later reinstating it once they had inspected the intermediate outputs. In one case, *constraint relieved or simplified* ($n = 1$, 9%) was more tool-focused. P9 simplified the prompt to make the target molecule simpler and reduce opportunities for scientific inaccuracies in the generated output. Thus, not all revisions provide the same kind of evidence for model refinement.

In contrast, a smaller subset of participants ($n = 3$, 27%) focused on *Rephrase*. These language-oriented changes involved substituting more precise terminology or elaborating descriptions to clarify intent without adding new chemical constraints. For example, P7 replaced a vague reference to a "base group" with the more specific term "hydrogen bond acceptor site," while P10 became more descriptive about the placement of AH, B, and X groups to better reflect the sweetness theory. Such revisions refined how existing ideas were communicated to GenAI. No instances of feedback-based or inquiry-based refinement were observed, which is consistent with the system's lack of responsive feedback.

Prompt structure evolution (CLEAR). At the prompt level, CLEAR analysis of all subsequent prompts ($n = 30$) showed that *Explicit* features remained dominant ($n = 27$, 90%). *Logical* structuring and *Adaptive* features each appeared in nearly half of subsequent prompts ($n = 14$, 47%), and *Restrictive* features appeared in fewer than half ($n = 12$, 40%). This distribution indicates that during revision, participants continued to use prompts as constraint-setting tools while adjusting the specificity, structure and boundaries of their requests. Changes within the Explicit dimension clarify how participants' prompts evolved from initial to subsequent iterations (Fig. 5 and 6), with illustrative excerpts for each subtheme provided in SI.

Although both the initial and subsequent prompts were coded with the same five subthemes, their distribution and richness shifted noticeably from first prompts to subsequent prompts, with *Functional Groups and Interaction Roles* emerging as a central focus. Within this dimension, functional group targeting was already central in first prompts ($n = 6$, 55%) and remained prominent in later prompts ($n = 13$, 43%). However, the nature of this explicitness became more mechanistic. In initial prompts, participants often named functional groups without fully specifying their interaction roles. In subsequent prompts, a larger proportion explicitly described how these groups should behave, with some referencing

Table 6 Participants' prompt revision strategies analyzed through REFINE

Theme	Excerpt	n (%)
(1) Rephrase		3 (27)
(1) Rephrasing language	"... contain a base group that could attract a hydrogen ion (P7, 3rd prompt) → "... contain a hydrogen bond acceptor site " (P7, 4th prompt)	2 (18)
(2) Being more descriptive	"I wanted to be even closer to AH-B-X model , emphasizing placement and all 3 groups. So I was more explicit about what the model required" (P10, 2nd strategy)	1 (9)
(2) Experiment with context, examples, constraints		9 (82)
(1) Constraint tightening	"This molecule contains a hydrophilic group -OH and a hydrophobic group -CF ₃ . But it is missing an electronegative atom to be the base to accept hydrogen atoms. So I added the last sentence to the prompt " (P5, 2nd strategy)	6 (55)
(2) Both tightening/relieving	"alter erythritol so it contains a benzene ring and a hydrogen bond acceptor site (tightening)" → "alter erythritol so it contains a benzene ring (relieving)" → " alter the previous molecule to contain an end that will accept a hydrogen bond (tightening)" (P7, 6-8th prompts)	2 (18)
(3) Constraint relieving/simplified	"I'll try revising the prompt to make the molecule simpler/smaller to see if that reduces the opportunities for scientific inaccuracies" (P9, 2nd strategy)	1 (9)
(3) Navigate by iterations		10 (91)
(4) Evaluate outputs		10 (91)

Note: percentages indicate the proportion of participants ($n = 11$) who used each strategy at least once. They do not represent how frequently a given strategy was repeated across prompts.

Explicit Dimension Subthemes (Subsequent Prompts)

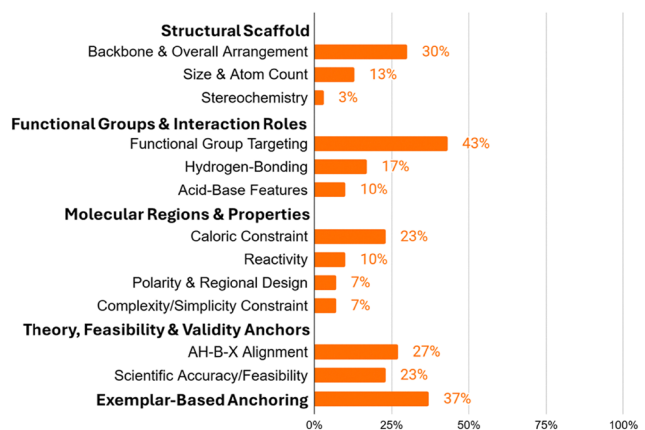


Fig. 5 Characteristics of subsequent prompts across explicit subthemes.

hydrogen-bonding roles ($n = 5$, 17%) or acid-base features ($n = 3$, 10%). For example, P7 required that the molecule includes both a hydrogen donor and a hydrogen bond acceptor and P9 requested to include an acidic hydrogen and a base, indicating the interaction roles they expected at the taste receptor. This pattern indicates a shift from naming which groups are to articulating what they should do in the context of sweetness.

A similar shift appeared in participants' attention to molecular structure and properties. For *Structural Scaffold*, the first prompts rarely specified the overall structure, with backbone or arrangement and size or atom count each appearing in one initial prompt ($n = 1$, 9%). In contrast, subsequent prompts foregrounded scaffold-related details more frequently: arrangement of functional groups ($n = 9$, 30%), size or atom-count

requirements ($n = 4$, 13%), and stereochemical constraints ($n = 1$, 3%). This shift illustrates that, over time, participants moved from simply ensuring that the right parts were present to specifying how those parts should be organized, the size of the molecule, and its orientation in space.

In terms of *Molecular Regions and Properties*, the first prompts focused almost exclusively on the caloric status, with noncaloric sweetener requirements appearing in two initial prompts ($n = 2$, 18%). In subsequent prompts, this category broadened and diversified. Caloric constraints appeared in seven subsequent prompts ($n = 7$, 23%), and new subthemes emerged, including limits on reactivity ($n = 3$, 10%), specifications of hydrophilic and hydrophobic regions ($n = 2$, 7%), and explicit expectations for overall complexity or simplicity ($n = 2$, 7%). In the later cycles, participants were not only stating that a molecule should be low or noncaloric but also treating the molecule as a system with multiple interacting regions and global properties such as polarity, reactivity, and structural complexity.

For *Theory, Feasibility, and Validity Anchors*, the references to the AH-B-X framework remained stable across cycles, appearing in both initial prompts ($n = 3$, 27%) and subsequent prompts ($n = 8$, 27%). In contrast, the explicit appeals to generic scientific accuracy or feasibility decreased from initial prompts ($n = 6$, 55%) to subsequent prompts ($n = 7$, 23%). Early prompts frequently included broad directives that molecules be "scientifically sound" or "accurate," whereas later prompts less often restated these broad constraints and instead encoded plausibility through detailed structural, functional, and regional specifications. This pattern demonstrates that, as participants iterated, they shifted from invoking plausibility as a general requirement to operationalizing it through more concrete

Percent Changes in Explicit Prompt Features from Initial vs Subsequent Prompts

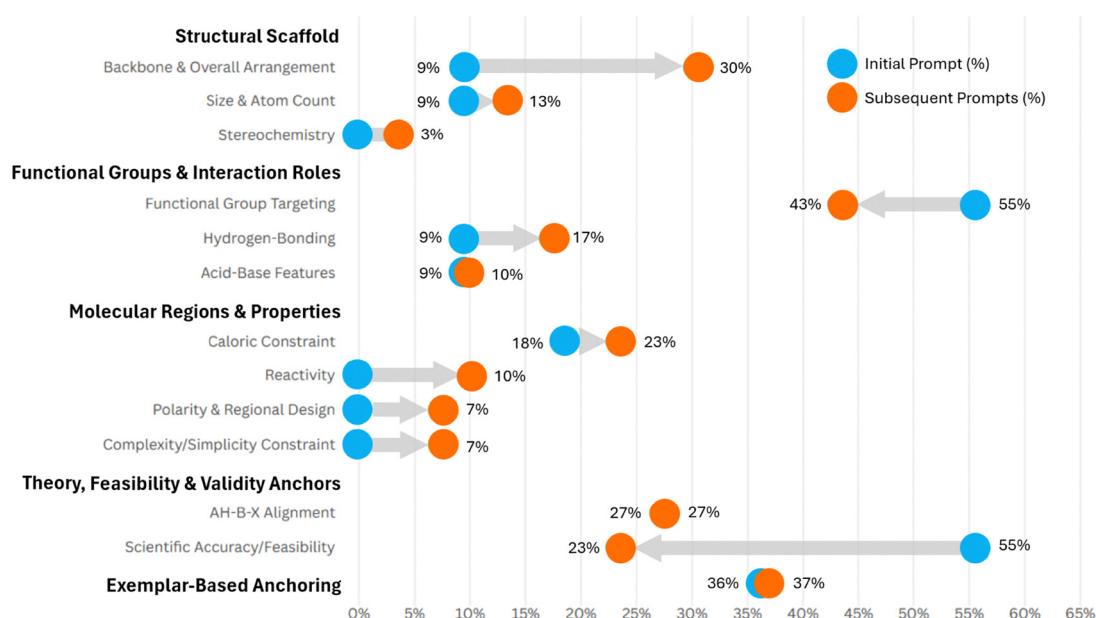


Fig. 6 Percentage changes in explicit prompt features from first to subsequent prompts.

design constraints. Within *Exemplar-Based Anchoring*, references to known sweeteners, such as aspartame or sucralose, appeared at similar levels in initial prompts ($n = 4$, 36%) and subsequent prompts ($n = 11$, 37%). This pattern suggests that exemplar-based anchoring remained a consistent strategy across prompt cycles.

Rationale for the final molecule

When justifying their final molecule selections, participants drew on theory-based and comparative reasoning (Table 7). The most common rationales referenced the AH-B-X theory ($n = 5$, 45%), particularly the presence of hydrogen donors, acceptors, and bulky hydrophobic groups, as well as the appropriate spatial arrangement among them. Other participants justified their selection by referencing the similarities to existing sweeteners, such as acesulfame potassium, sucralose, or neotame ($n = 3$, 27%), positioning their choice as credible by comparison. Less frequently, participants emphasized chemical properties, pointing to specific features such as hydrogen acceptor capacity, interactions with taste receptors, or provided no specific rationale (each $n = 1$, 9%). Notably, some participants were unable to provide a rationale due to unsatisfactory outputs, as their prompts failed to generate a molecule that met their design intent ($n = 4$, 36%).

Participants' final sweetener selections reflected reasoned modeling decisions, not an attempt to identify a correct or chemically validated solution. When participants accepted a candidate molecule, they generally did so by referencing alignment with their design goals, theoretical criteria, or comparative anchors. When they rejected or failed to justify a molecule, this also reflected the limits of the generated outputs and the uncertainty involved in judging molecular plausibility within the task.

Thus, final molecule selection functioned as evaluative judgment under uncertainty. Participants treated GenAI outputs as candidate, provisional representations that require interpretation and justification, with acceptance depending on how well each candidate aligned with structural, theoretical, and functional criteria.

Discussion

This study contributes to chemistry education research by characterizing the potential of GenAI as a mediating tool within

modeling practice rather than as a source of answers or representations to be accepted at face value. Across participants, engagement with “a chemistry-specific GenAI-supported modeling tool” was organized according to modeling goals, evaluative criteria, and iterative refinement, with output molecules treated as provisional candidates requiring interpretation and justification. Building on recent work on GenAI in chemistry education (Tassoti, 2024; Akaygun and Kilic, 2025) and prompt-engineering strategies for educators (Park and Choo, 2025), the study strengthens this conversation by situating prompt-based interaction within scientific modeling. The integration of GenAI into participants' workflows was heavily guided by theoretical frameworks and disciplinary boundaries, which effectively transformed prompt construction, output assessment, and iterative modification into extensions of scientific modeling practice.

However, the efficacy of this approach varied among individuals. As detailed in Table 7, 36% of participants could not form a clear rationale because they received unsatisfactory outputs, a finding that introduces nuance into the overall assertion that GenAI facilitated modeling throughout the study. In successful instances, participants applied rigorous theory-driven metrics; for example, P4 noted that a generated structure “Meets AH-B-X structure” due to the accurate spacing of its respective groups, while participant P1 successfully mapped out the three necessary requirements as “an amine group as H donor, =O group from ester as H acceptor, and benzene group as hydrophobic bulky group”. Conversely, four participants were unable to arrive at such justificatory conclusions because the platform failed to yield a molecular structure that aligned with their initial design objectives. When participants were able to generate interpretable outputs, evaluate those outputs against chemical and theoretical criteria, and continue revising within the available time and scaffolding, GenAI proved capable of supporting modeling practice collaboratively.

By emphasizing goal-directed design, criteria-based assessment, and navigation through ambiguous outcomes, these insights broaden current frameworks of molecular modeling in chemistry education by validating GenAI as a legitimate representational asset for learners. This framing shifts attention from the correctness of AI outputs to the quality of learners' modeling practices, offering a principled way to integrate GenAI into chemistry instruction without undermining disciplinary rigor. This perspective is particularly relevant given current research on emerging technology adoption

Table 7 Types of rationales used by participants to justify their final molecule selections

Theme	Excerpt	<i>n</i> (%)
(1) Theory reference	“Meets AH-B-X structure –AH: –NH-B: C=O-X: aromatic ring” (P4)	5 (45)
(2) Reference to existing sweeteners	“... and possesses similarities to acesulfame potassium ” (P3)	3 (27)
(3) Chemical properties	“... they might be better hydrogen acceptors ...” (P2)	1 (9)
(4) Interaction with taste receptors	“... it has an amine group as H donor, =O group from ester as H acceptor, and benzene group as hydrophobic bulky group to interact with taste receptors ” (P1)	1 (9)
(5) No specific reasons	“Yes, I think this molecule fits the model, I feel like it read my mind by taking out the Fluorines” (P10)	1 (9)
(6) N/A (failed generating a desired molecule)		4 (36)

Note: percentages indicate the proportion of participants ($n = 11$) who used each type of rationale at least once.

among teachers, which highlights the importance of perceived usefulness, compatibility, and contextual support in shaping teachers' intentions to use new technologies (Frawley and Campbell, 2025). Instead of treating GenAI adoption as a matter of access or efficiency alone, this study shows how GenAI can become instructionally meaningful when embedded in a disciplinary task that requires teachers to articulate, test, and refine chemical ideas.

Grounded in Modeling Theory and guided by the CLEAR and REFINE frameworks (Lo, 2023; Park and Choo, 2025), the findings indicate that participants enacted core epistemic practices of model construction, evaluation, revision, and justification. These practices illustrate how conceptual understanding can be externalized, tested, and refined in a generative environment, consistent with prior work emphasizing modeling as a central epistemic process in science education (Gilbert, 2004; NGSS Lead States, 2013).

These findings should also be understood in relation to the instructional design of the task. Participants had already engaged with the AH-B-X theory of sweetness and molecular structure–function relationships, and the worksheet organized their work around prompts, outputs, evaluations, and revision strategies. These scaffolds likely supported participants in treating GenAI outputs as candidates to be evaluated and refined. The following sections synthesize findings across research questions.

Goals, prompt construction, and iterative refinement

Participants translated their disciplinary knowledge into natural language prompts when constructing initial models. Their stated design goals suggest that they viewed prompt construction as both a way to encode chemical knowledge and a way to interact strategically with GenAI. Most initial prompts were explicit and adaptive, specifying functional groups, atom counts, or reference sweeteners, and many were anchored in the AH-B-X theory (Zheng *et al.*, 2024). In MLC, this work aligns with early model specification in the Model Adduction phase, where prompts operate as written models about structure–function relationships that may yield sweetness. Participants therefore approached prompt construction as defining the parameters and plausibility space of a candidate model.

This finding extends recent work on prompt engineering by showing how prompt engineering can function as disciplinary modeling in a chemistry context. Lo (2023) and Park and Choo (2025) frame prompt engineering as an emerging literacy, while Tassoti (2024) emphasizes the need to examine students' prompt engineering strategies in chemistry education. Building on this work, this study demonstrates that prompt quality in a molecular design task was not just about improving AI outputs. Prompts functioned as observable artifacts that provided evidence of participants' chemical reasoning, encoding assumptions about molecular structure, functional groups, spatial arrangements, and sweetness-related mechanisms. Therefore, prompt engineering can serve as a form of modeling where participants represented and refined their ideas with GenAI, expanding on prior findings

that constructing and refining models fosters conceptual understanding (Bamberger and Davis, 2013; Forbes *et al.*, 2015).

Participants' evaluation practices exemplify model testing and further reinforce a modeling orientation. The generated molecules were evaluated by examining chemical properties, assessing theoretical alignment with the AH-B-X, and verifying the scientific accuracy and plausibility. These activities align with the evaluation phase of MLC, where learners test the adequacy of a model against theoretical and empirical expectations. Participants coordinated their conceptual understanding with the AI-generated outputs, treating these outputs as models to be examined. This process illustrates how GenAI-supported modeling can expand opportunities for reasoning while reinforcing the need for disciplined evaluation grounded in theory and evidence. This echoes previous work that highlights the cognitive and representational demands of chemistry modeling (Gkitzia *et al.*, 2020; Talanquer, 2022). Such evaluative practices provided the foundation for subsequent refinement, as participants revised their prompts to improve the quality and plausibility of their generated molecules.

Prompt revisions reflected purposeful and iterative refinement, although some revisions may also reflect participants' adaptation to the GenAI tool. Over time, revisions shifted from broad constraints ("scientifically accurate," "noncaloric") toward local control over specific features of the molecule, including functional roles, spatial arrangements, and reactivity. This trajectory suggests a transition from broad Model Adduction to fine-grained Model Formulation. Once GenAI's default behaviors and limitations became more visible, participants responded by refining the model more explicitly focused on structure and function. Iteration with GenAI thus created a context in which participants had to unpack their conceptual understanding of sweetness and molecular structure into explicit design moves, instead of relying on generic feasibility language.

Analyses of the explicit dimension further show that this refinement involved increasing mechanistic specificity. While first prompts included structure–function related constraints, AH-B-X references, or noncaloric status, these references often functioned as high-level labels. Subsequent prompts increasingly encoded mechanistic details, such as which groups should donate or accept hydrogen, how hydrophobic bulk should be positioned, and how distinct regions of the molecule should interact. At the same time, explicit appeals to generic scientific accuracy declined as participants began to operationalize plausibility through concrete structural, functional, and regional constraints. For example, the request to *be scientifically accurate* was progressively translated into requirements on stereochemistry, hydrogen-bonding capacity, reactivity limits, and polarity patterns, with plausibility checks occurring both inside the prompt (*via* detailed constraints) and outside it (*via* written evaluations and decisions to accept or reject outputs). This pattern indicates that participants viewed revision as a deliberate process of improving AI-generated molecules through systematic experimentation, similar to how students refine models through empirical testing in modeling-based learning (Roth *et al.*, 2020; Malone and Schuchardt, 2023).

Justification and selection of molecules

Participants' final justifications for molecule selection reflected both theory-based and comparative reasoning. They frequently cited alignment with the AH-B-X theory (Zheng *et al.*, 2024), emphasizing correct spatial and functional relationships among molecular components. Others drew comparisons to known sweeteners or withheld acceptance when outputs failed to meet theoretical expectations or representational accuracy. Importantly, 36% of participants could not obtain an output that satisfied their design goals. The finding may indicate that some participants did not fully engage in the iterative modeling process, opting to discontinue the task rather than continue refining prompts, testing outputs, and revising constraints until a viable molecule was produced.

This finding may also reflect the technology's inability to address participants' intended design goals or requests, which is plausible given that this was an initial version of the GenAI system. Successful use of the system may therefore depend not only on the capabilities of the GenAI tool itself but also on participants' prompt-engineering skills, familiarity with collaborating with GenAI systems, and sufficient time to engage in iterative cycles of modeling and refinement. This interpretation is important because productive modeling often depends on sustained processes of evaluation and revision, particularly when generated outputs do not initially align with expectations.

The absence of a final selection should not be interpreted only as task failure. Withholding acceptance could reflect critical evaluation and epistemic responsibility, as participants recognized that generated molecules did not meet the criteria. This resonates with the view that effective modeling involves not only accepting and refining models but also rejecting models that lack coherence or adequacy (Halloun, 2006). Thus, final molecule selection, or non-selection, represented evaluative judgment under uncertainty.

Within this bounded case, participants primarily engaged GenAI as a modeling amplifier. GenAI broadened the space of possible molecular representations, while participants provided theoretical coherence, constraint discipline, and critical evaluation. This clarifies how generative systems can support, rather than replace, human reasoning. AI rapidly generates structural possibilities, whereas learners ensure mechanistic plausibility and theoretical alignment. The resulting partnership illustrates that meaningful learning occurs not in accepting AI outputs but in the iterative dialogue between human reasoning and computational generation. In this way, the study connects recent work on GenAI prompt engineering and representation with modeling, offering a framework for using GenAI to make chemical reasoning more visible, assessable, and instructionally productive.

Implications

This study extends modeling theory by clarifying how established phases manifest when a generative system is positioned as a co-modeler. Prompt creation represents model construction,

evaluation corresponds to model testing, revision serves as experimental manipulation, and justification becomes epistemic validation. These reinterpretations illustrate that GenAI-supported modeling maintains the structure of MLC while introducing new forms of coordination between humans and AI.

The study also suggests the value of text-based artifacts for examining prompt-based modeling. Prompt logs, evaluation notes, revision notes, and justifications provided useful access to how participants articulated, tested, and refined chemical constraints across iterative cycles. The combined use of CLEAR and REFINE helped capture both the structure of prompts and the ways those prompts evolved over time. These methodological approaches offered a systematic way to trace key aspects of their disciplinary reasoning in an AI-supported environment.

Consequently, these findings suggest the educational potential of chemistry-specific GenAI applications when implemented as a provisional modeling partner whose outputs remain subject to human judgment. In this study, participants had to evaluate chemical plausibility, theoretical alignment, and adequacy of fit. This suggests that the educational value of GenAI in molecular modeling lies less in automated generation itself than in how it supports iterative evaluation and revision grounded in disciplinary reasoning.

Limitations and future research directions

Several limitations should be acknowledged. The study's sample and context limit the transferability of the findings. The study involved a small sample of experienced secondary teachers participating in a bounded professional learning context, and their disciplinary expertise and pedagogical orientations likely shaped how they engaged in prompt-based modeling. Accordingly, the findings should not be generalized to students, pre-service teachers, or broader teacher populations without further study, nor should they be interpreted as direct evidence of how GenAI-supported modeling would unfold in classroom settings. Future research with larger and more diverse samples is needed to identify broader patterns and variations in these practices.

The GenAI-supported modeling tool did not always produce chemically valid structures, requiring participants to draw on prior chemical knowledge to evaluate the generated outputs. The platform also did not show GenAI's reasoning to users, preventing recursive exchanges that could have supported feedback loops and deeper refinement. Because iterative dialogue is central to model-based reasoning, future research should examine how displaying reasoning in GenAI systems influences the continuity, depth, and epistemic quality of learners' revisions.

The study relied on text-based artifacts, which, while analytically rich, provide limited insight into participants' moment-to-moment cognitive processes and cannot verify whether prompts were authored independently or created with assistance from another GenAI system. Complementary methods such as think-aloud protocols or screen-capture analyses could

better document both prompt authorship and how reasoning unfolds in real time during AI-human interaction.

The study also highlights a limitation of current language models in GenAI-supported modeling environments. Although the LLM could generate structurally rich candidate molecules, it could not determine whether those structures were chemically plausible or theoretically well aligned, leaving that evaluative work to participants. Future research should therefore examine how GenAI-supported systems can make this evaluative work more explicit and better support human judgment by making prompt evolution, constraint changes, and resulting structural shifts more visible to users.

Conclusion

This study demonstrates that GenAI can be productively integrated into chemistry education when it is positioned as a mediating tool within modeling practice. Experienced secondary teachers engaged a chemistry-specific GenAI-supported modeling tool through design goals, criteria-based evaluation, and iterative refinement, treating AI-generated molecules as provisional representations that required interpretation, justification, and revision. Across cases, prompt construction, evaluation, and selection functioned as extensions of disciplinary modeling work.

By documenting how participants constrained, evaluated, and justified AI-generated molecules using theoretical and structural criteria, this study extends existing accounts of modeling in chemistry education to include generative tools as part of learners' modeling resources. Importantly, modeling decisions were made under acknowledged uncertainty, with final selections framed as reasoned judgments. This orientation preserves disciplinary rigor while leveraging GenAI's capacity to expand the space of candidate representations. Looking forward, the findings suggest that the educational value of GenAI lies in how it is embedded within tasks that foreground modeling goals, theoretical constraints, and evaluative reasoning. When designed in this way, GenAI has the potential to amplify core epistemic practices in chemistry learning while keeping human judgment firmly at the center.

Conflicts of interest

The authors report there are no competing interests to declare.

Ethical approval

This research was conducted in compliance with ethical guidelines and received approval from the University of Florida Institutional Review Board (IRB No. 202400747).

Consent to participate

Formal consent was secured from the participants. To ensure confidentiality and protect their privacy, all data reported in the study was anonymized.

Data availability

The research data supporting this article are not publicly available because participants did not consent to public sharing of their data and public release could compromise participant privacy. De-identified excerpts and coding summaries supporting the analysis are provided in the article and supplementary information (SI). Supplementary information is available. See DOI: <https://doi.org/10.1039/d6rp00064a>.

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References

- Akaygun, S. and Kilic, I., (2025), Generative artificial intelligence (GenAI) as the artist of chemistry visuals: chemistry preservice teachers' reflections on visuals created by GenAI, *J. Chem. Educ.*, **102**(7), 2549–2564, DOI: [10.1021/acs.jchemed.4c00775](https://doi.org/10.1021/acs.jchemed.4c00775).
- Bamberger, Y. M. and Davis, E. A., (2013), Middle-School Science Students' Scientific Modelling Performances Across Content Areas and Within a Learning Progression, *Int. J. Sci. Educ.*, **35**(2), 213–238, DOI: [10.1080/09500693.2011.624133](https://doi.org/10.1080/09500693.2011.624133).
- Bereczki, E. O. and Kárpáti, A., (2021), Technology-enhanced creativity: A multiple case study of digital technology-integration expert teachers' beliefs and practices, *Think. Skills Creat.*, **39**, 100791, DOI: [10.1016/j.tsc.2021.100791](https://doi.org/10.1016/j.tsc.2021.100791).
- Berg, A., Orraryd, D., Pettersson, A. J. and Hultén, M., (2019), Representational challenges in animated chemistry: Self-generated animations as a means to encourage students' reflections on sub-micro processes in laboratory exercises, *Chem. Educ. Res. Pract.*, **20**, 710–737, DOI: [10.1039/C8RP00288F](https://doi.org/10.1039/C8RP00288F).
- Braun, V. and Clarke, V., (2006), Using thematic analysis in psychology, *Qual. Res. Psychol.*, **3**(2), 77–101, DOI: [10.1191/1478088706qp063oa](https://doi.org/10.1191/1478088706qp063oa).
- Clark, V. P. and Creswell, J. W., (2015), *Understanding research: A consumer's guide*.
- Forbes C., Vo T., Zangori L. and Schwarz C., (2015), Using Models Scientifically: Scientific Models help Students Understand the Water Cycle, *Sci. Child.*, **53**(2), 42–49, DOI: [10.2505/4/sc15_053_02_42](https://doi.org/10.2505/4/sc15_053_02_42).
- Frawley, C. and Campbell, L. O., (2025), Factors that predict teachers' intentions to utilize emerging technologies: An investigation using PLS-SEM, *Educ. Inf. Technol.*, **30**(2), 1589–1606, DOI: [10.1007/s10639-024-12796-8](https://doi.org/10.1007/s10639-024-12796-8).
- Gilbert, J. K., (2004), Models and Modelling: Routes to More Authentic Science Education, *Int. J. Sci. Math. Educ.*, **2**(2), 115–130, DOI: [10.1007/s10763-004-3186-4](https://doi.org/10.1007/s10763-004-3186-4).
- Gkitzia, V., Salta, K. and Tzougraki, C., (2020), Students' competence in translating between different types of chemical

- representations, *Chem. Educ. Res. Pract.*, **21**, 307–330, DOI: [10.1039/c8rp00301g](https://doi.org/10.1039/c8rp00301g).
- Halloun, I. A., (2006), *Modeling theory in science education*, Springer Netherlands.
- Hestenes, D., (2010), Modeling theory for math and science education, in R. Lesh, P. L. Galbraith, C. R. Haines and A. Hurford (ed.), *Modeling students' mathematical modeling competencies*, Springer US, pp. 13–41.
- Institute for Future Intelligence, (2025), *AIMS: Artificial Intelligence for Molecular Sciences*, Retrieved August 9, 2025, from <https://intofuture.org/aims.html>.
- Lo, L. S., (2023), The art and science of prompt engineering: A new literacy in the information age, *Internet Ref. Serv. Q.*, **27**(4), 203–210, DOI: [10.1080/10875301.2023.2227621](https://doi.org/10.1080/10875301.2023.2227621).
- Malone, K. L. and Schuchardt, A., (2023), Modelling-based pedagogy as a theme across science disciplines-Effects on scientific reasoning and content understanding, *Eur. J. Sci. Math. Educ.*, **11**(4), 717–737, DOI: [10.30935/scimath/13516](https://doi.org/10.30935/scimath/13516).
- McHugh, M. L., (2012), Interrater reliability: The kappa statistic, *Biochem. Med.*, **22**(3), 276–282, DOI: [10.11613/BM.2012.031](https://doi.org/10.11613/BM.2012.031).
- NGSS Lead States, (2013), *Next generation science standards: For states, by states*, The National Academies Press.
- O'Connor, C. and Joffe, H., (2020), Intercoder reliability in qualitative research: debates and practical guidelines, *Int. J. Qual. Methods*, **19**, 1–13, DOI: [10.1177/1609406919899220](https://doi.org/10.1177/1609406919899220).
- Padalkar, S. and Hegarty, M., (2015), Models as feedback: Developing representational competence in chemistry, *J. Educ. Psychol.*, **107**(2), 451–467, DOI: [10.1037/a0037516](https://doi.org/10.1037/a0037516).
- Palinkas, L. A., Horwitz, S. M., Green, C. A., Wisdom, J. P., Duan, N. and Hoagwood, K., (2015), Purposeful sampling for qualitative data collection and analysis in mixed method implementation research, *Administrat. Pol. Mental Health*, **42**(5), 533–544, DOI: [10.1007/s10488-013-0528-y](https://doi.org/10.1007/s10488-013-0528-y).
- Park, J. and Choo, S., (2025), Generative AI prompt engineering for educators: practical strategies, *J. Spec. Educ. Technol.*, **40**(3), 411–417, DOI: [10.1177/01626434241298954](https://doi.org/10.1177/01626434241298954).
- Roth, T., Scharfenberg, F.-J., Mierdel, J. and Bogner, F. X., (2020), Self-evaluative Scientific Modeling in an Outreach Gene Technology Laboratory, *J. Sci. Educ. Technol.*, **29**(6), 725–739, DOI: [10.1007/s10956-020-09848-2](https://doi.org/10.1007/s10956-020-09848-2).
- Stieff, M., Scopelitis, S., Lira, M. E. and Desutter, D., (2016), Improving Representational Competence with Concrete Models, *Sci. Educ.*, **100**(2), 344–363, DOI: [10.1002/sce.21203](https://doi.org/10.1002/sce.21203).
- Talanquer, V., (2022), The Complexity of Reasoning about and with Chemical Representations, *JACS Au*, **2**(12), 2658–2669, DOI: [10.1021/jacsau.2c00498](https://doi.org/10.1021/jacsau.2c00498).
- Tassoti, S., (2024), Assessment of students use of generative artificial intelligence: prompting strategies and prompt engineering in chemistry education, *J. Chem. Educ.*, **101**(6), 2475–2482, DOI: [10.1021/acs.jchemed.4c00212](https://doi.org/10.1021/acs.jchemed.4c00212).
- VERBI Software, (2024), *MAXQDA 24.2 [computer software]*, <https://maxqda.com>.
- Zheng, H., Xu, X., Fang, Y., Sun, R. and Liu, B., (2024), The molecular theory of sweet taste: revisit, update, and beyond, *J. Med. Chem.*, **67**(5), 3232–3243, DOI: [10.1021/acs.jmedchem.3c02055](https://doi.org/10.1021/acs.jmedchem.3c02055).